



GIRRAWEEEN HIGH SCHOOL
HALF YEARLY EXAMINATIONS
2018
MATHEMATICS
EXTENSION 2

*Time Allowed: Two hours
(Plus 5 minutes reading time)*

Total Marks: 95

Instructions:

- There are 11 questions in this paper. All questions are compulsory.
- Use blue or black pen.
- Write all your answers in the Answer Booklets provided.
- For Questions 1 – 5, fill in the circle corresponding to the correct answer in your answer booklet.
- For Questions 6 – 11, start each question on a new page.
- Write on both sides of the paper.
- Show all necessary working.
- Board-approved calculators may be used.
- Mathematics reference sheets are provided.
- Marks may be deducted for careless or badly arranged work.
- Write 'End of Solutions' at the conclusion of your solutions to the task.

Questions 1 - 5 (5 marks)

Fill in the circle corresponding to the correct answer in your answer booklet.

1 What is the value of $\frac{6}{iz}$ if $z = -1 + i$?

(A) $-3 - 3i$

(B) $-3 + 3i$

(C) $3 - 3i$

(D) $3 + 3i$

2 If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^{2017}$ is equal to:

(A) $-2^{2017}\omega$

(B) $2^{2017}\omega$

(C) $-2^{2017}\omega^2$

(D) $2^{2017}\omega^2$

3 The equation $24x^3 - 12x^2 - 6x + 1 = 0$ has roots α, β and γ .
What is the value of α if $\alpha = \beta + \gamma$.

(A) $-\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 1

4 The equations of the asymptotes of the hyperbola $9x^2 - 4y^2 = 36$ are:

(A) $x = \pm \frac{3}{2}y$

(B) $y = \pm \frac{3}{2}x$

(C) $x = \pm \frac{9}{4}y$

(D) $y = \pm \frac{9}{4}x$

5 The foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$ are:

(A) $(0, \pm \sqrt{7})$

(B) $(\pm \sqrt{7}, 0)$

(C) $\left(0, \pm \frac{\sqrt{7}}{4}\right)$

(D) $\left(\pm \frac{\sqrt{7}}{4}, 0\right)$

Question 6 (28 marks)

a. Let $\omega_1 = 8 - 2i$ and $\omega_2 = -5 + 3i$. Find the value of:

[2]

(i) $\omega_1 \omega_2$

(ii) $\omega_1 + \overline{\omega_2}$

[2]

b. Reduce the complex expression $\frac{(2-i)(8+3i)}{3+i}$ to the form $a + ib$ where a and b are real numbers.

[3]

c. (i) Express $\sqrt{3} - i$ in modulus-argument form.

[2]

(ii) Hence evaluate $(\sqrt{3} - i)^6$.

[3]

d. Sketch the region defined by:

$$|z - (3 + i)| \leq 3 \text{ and } \frac{\pi}{4} < \arg [z - (1 + i)] \leq \frac{\pi}{2} \text{ are satisfied.}$$

[4]

e. If $z = \cos \theta + i \sin \theta$

(i) Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$.

[2]

(ii) Hence show that $\cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$.

[3]

f. (i) Show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$.

[2]

(ii) Using the equation $8x^3 - 6x - 1 = 0$ and letting $x = \cos \theta$,

[1]

$$\text{deduce that } \cos 3\theta = \frac{1}{2}.$$

(iii) Find the roots of $8x^3 - 6x - 1 = 0$ in terms of $\cos \theta$.

[2]

(iv) Hence evaluate $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$.

[2]

Question 7 (20 marks)

a. The equation $x^3 - 4x^2 + 5x + 2 = 0$ has roots α, β and γ .

Find (i) $\alpha + \beta + \gamma$ [1]

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ [1]

(iii) $\alpha^2 + \beta^2 + \gamma^2$ [2]

(iv) $(2 + \alpha)(2 + \beta)(2 + \gamma)$ [2]

b. (i) Find a and b such that $x = 2$ is a double root of $f(x) = x^4 + ax^3 + x^2 + b$. [2]

(ii) For the values of a and b above, factorise $f(x)$ over the real numbers. [2]

c. Let α, β and γ be the roots of the equation $x^3 - 5x^2 + 5 = 0$.

(i) Find a polynomial equation with integer coefficients whose roots are $\alpha - 1, \beta - 1$ and $\gamma - 1$. [2]

(ii) Find a polynomial equation with integer coefficients whose roots are α^2, β^2 and γ^2 . [2]

(iii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$. [2]

d. Given that $2 + 3i$ is a solution to $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$

(i) State why $2 - 3i$ is also a solution. [1]

(ii) Hence, completely solve $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$ [3]

Question 8 (12 marks)

a. Point $P(x_0, y_0)$ is on the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

(i) Find the coordinates of the foci and the equations of the directrices. [2]

(ii) Sketch the ellipse, clearly showing all important features. [2]

(iii) Show that the equation of the tangent at P is $\frac{xx_0}{25} + \frac{yy_0}{9} = 1$. [2]

(iv) Let the tangent at P meet a directrix at a point Q .

Show that $\angle PSQ$ is a right angle, where S is the corresponding focus. [4]

b. Describe the locus of z , stating its equation, if $|z + 3| + |z - 3| = 10$. [2]

Question 9 (7 marks)

Consider the hyperbola $\frac{x^2}{4} - \frac{y^2}{12} = 1$.

(i) Sketch the graph of the hyperbola, clearly showing:
- any intercepts made with the axes
- the coordinates of the foci
- the equations of the directrices
- the equations of the asymptotes. [4]

(ii) $T(x_1, y_1)$ is a point on the hyperbola.

The equation of the tangent to the hyperbola at T is

$$\frac{xx_1}{4} - \frac{yy_1}{12} = 1 \text{ [Do not prove this].}$$

$P(4, 6)$ and $Q(14, 24)$ are two points on the hyperbola.

M is the midpoint of PQ and O is the origin.

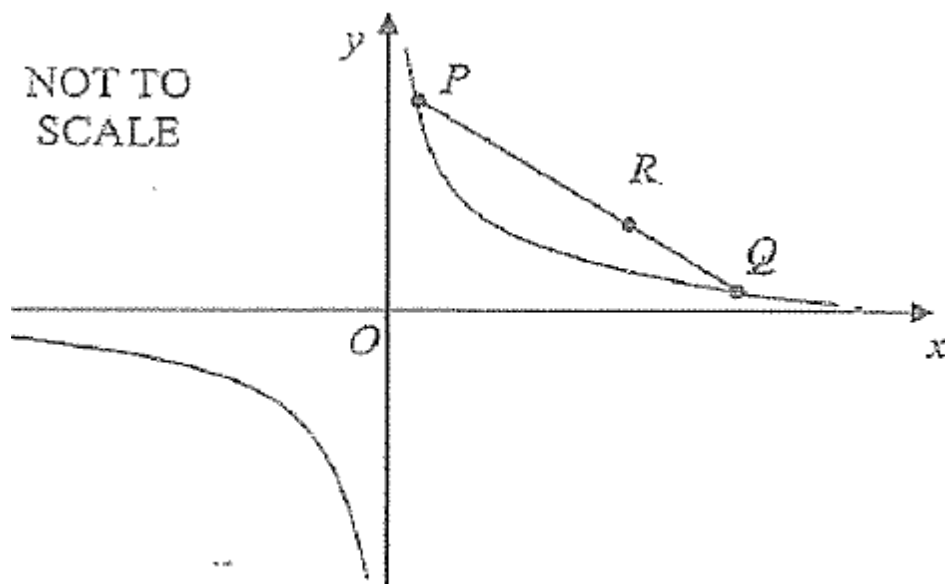
The tangents to the hyperbola at P and Q intersect at the point R .

Show that the points R , O and M are collinear. [3]

Question 10 (15 marks)

- a. $P\left(2p, \frac{2}{p}\right)$ and $Q\left(2q, \frac{2}{q}\right)$ are points on the rectangular hyperbola $xy = 4$.

M is the midpoint of the chord PQ . P and Q move on the hyperbola so that the chord PQ always passes through the point $R(4, 2)$.



- (i) Show that the equation of the chord PQ is $x + pqy = 2(p + q)$. [3]

- (ii) Show that $pq = p + q - 2$. [1]

- (iii) Hence find the equation of the locus of M , as P and Q move on the curve $xy = 4$. [3]

- b. $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$.

Tangents to the hyperbola at P and Q intersect at the point R .

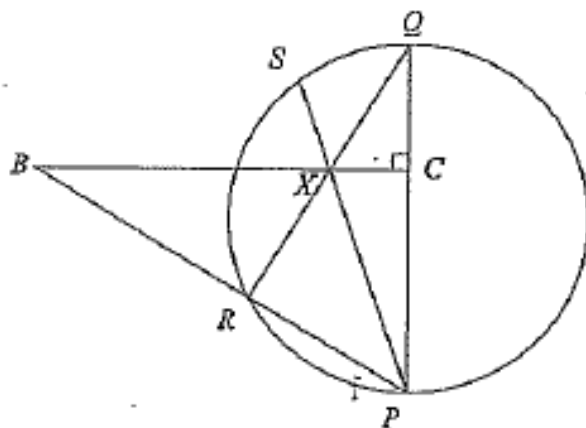
- (i) Show that the tangent to the rectangular hyperbola at the point $\left(ct, \frac{c}{t}\right)$ has the equation $x + t^2y = 2ct$. [2]

- (ii) Show that $R = \left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ [3]

- (iii) If P and Q are variable points on the rectangular hyperbola which move so that $p^2 + q^2 = 2$, find the equation of the locus of R . [3]

Question 11 (8 marks)

In the diagram, PQ is a diameter of the circle. The chords QR and PS intersect at X . The point C lies on PQ such that XC is perpendicular to PQ . The point B is the intersection of PR produced and CX produced.



- (i) Trace or copy the diagram into your booklet and show that $\angle PBC = \angle PQR$. [2]
- (ii) Show that $SBRX$ is a cyclic quadrilateral. [3]
- (iii) Show that the points B , S and Q are collinear. [3]

END OF EXAMINATION

EXTENSION 2
HY 2018 SOLUTIONS

MC

$$1. \frac{6}{iz} = \frac{6}{i(-1+i)} = \frac{6}{-1-i} \cdot \frac{-1+i}{-1+i}$$

$$= \frac{6(-1+i)}{2}$$

$$= -3 + 3i \quad \boxed{B}$$

$$2. (1 + \omega - \omega^2)^{2017}$$

$$= (1 + \omega + \omega^2 - 2\omega^2)^{2017}$$

$$= (-2\omega^2)^{2017} \quad [1 + \omega + \omega^2 = 0]$$

$$= (-2)^{2017} \cdot \omega^{4034}$$

$$= (-2)^{2017} \cdot \omega^2 \cdot \omega^{4032}$$

$$= (-2)^{2017} \cdot \omega^2 \cdot (\omega^3)^{1344} \quad [\omega^3 = 1]$$

$$= -2 \omega^2 \quad \boxed{C}$$

$$3. \text{Sum of roots: } \alpha + \beta + \gamma = \frac{-(-12)}{24} = \frac{1}{2}$$

$$\alpha + \alpha = \frac{1}{2}$$

$$2\alpha = \frac{1}{2}$$

$$\alpha = \frac{1}{4}$$

$\boxed{B}$

$$4. 9x^2 - 4y^2 = 36$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

$$a=2, b=3$$

$$\text{Asymptotes: } y = \pm \frac{b}{a}x$$

$$y = \pm \frac{3}{2}x$$

$\boxed{B}$

$$5. \frac{x^2}{9} + \frac{y^2}{16} = 1$$

$$a=3, b=4$$

major axis along y-axis

$$\text{Foci } (0, \pm be)$$

$$= (0, \pm \sqrt{7})$$

$\boxed{A}$

$$e^2 = 1 - \frac{a^2}{b^2}$$

$$= \frac{7}{16}$$

$$e = \frac{\sqrt{7}}{4}$$

Question 6

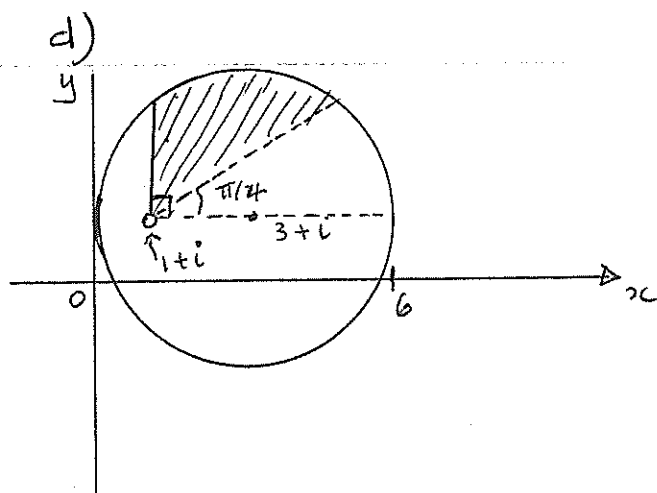
a) $w_1 = 8 - 2i$; $w_2 = -5 + 3i$

(i) $w_1 w_2$
 $= (8 - 2i)(-5 + 3i)$
 $= -40 + 24i + 10i + 6$
 $= -34 + 34i$

(ii) $w_1 + \overline{w_2}$
 $= 8 - 2i - 5 - 3i$
 $= 3 - 5i$

b) $\frac{(2-i)(8+3i)}{3+i}$
 $= \frac{16 + 6i - 8i + 3}{3+i}$
 $= \frac{19 - 2i}{3+i} \times \frac{3-i}{3-i}$
 $= \frac{55 - 25i}{10}$
 $= \frac{11}{2} - \frac{5i}{2}$

c) (i) $\sqrt{3} - i$
 $|\sqrt{3} - i| = \sqrt{3+1} = 2$
 $\arg(\sqrt{3} - i) = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$
 $= -\frac{\pi}{6}$
 $\sqrt{3} - i = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right)$
 (ii) $(\sqrt{3} - i)^6 = \left(2 \operatorname{cis}\left(-\frac{\pi}{6}\right)\right)^6$
 $= 64 \operatorname{cis}(-\pi)$
 $= 64(\cos(-\pi) + i \sin(-\pi))$
 $= -64$



e) (i) $z = \cos \theta + i \sin \theta$
 $z^n = \cos n\theta + i \sin n\theta$ (De Moivre's thm)
 $z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$
 $= \cos n\theta - i \sin n\theta$ (sin odd, cos even)
 $\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta$

(ii) $\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$
 $(2 \cos \theta)^4 = \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$

$$16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$$

$$\cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$$

f) i) $\cos 3\theta + i \sin 3\theta$
 $= (\cos \theta + i \sin \theta)^3$
 $= \cos^3 \theta + 3i \sin \theta \cos^2 \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$

Equating real parts,

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta \\ &= 4 \cos^3 \theta - 3 \cos \theta \end{aligned}$$

$$f \text{ (ii)} \quad 8x^3 - 6x - 1 = 0$$

$$\Rightarrow 4x^3 - 3x = \frac{1}{2}$$

$$\text{Let } x = \cos \theta$$

$$4\cos^3 \theta - 3\cos \theta = \frac{1}{2}$$

$$\therefore \cos 3\theta = \frac{1}{2}$$

$$\text{(iii)} \quad \cos 3\theta = \frac{1}{2}$$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$$\therefore x = \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$$

$$= \cos \frac{\pi}{9}, -\cos \frac{4\pi}{9}, -\cos \frac{2\pi}{9}$$

iv) Product of roots :

$$\left(\cos \frac{\pi}{9}\right) \left(-\cos \frac{4\pi}{9}\right) \left(-\cos \frac{2\pi}{9}\right) = \frac{1}{8}$$

$$\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$$

Question 7

$$a. \quad x^3 - 4x^2 + 5x + 2 = 0$$

$$i) \quad \alpha + \beta + \gamma = 4$$

$$ii) \quad \alpha\beta + \beta\gamma + \gamma\alpha = 5$$

$$\begin{aligned} iii) \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= 4^2 - 2(5) \\ &= 6 \end{aligned}$$

$$iv) \quad (2 + \alpha)(2 + \beta)(2 + \gamma)$$

$$= 8 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) + 4(\alpha + \beta + \gamma) + \alpha\beta\gamma$$

$$= 8 + 2(5) + 4(4) + (-2)$$

$$= 32$$

Q7(cont)

b. (i) $f(x) = x^4 + ax^3 + x^2 + b$

$$f'(x) = 4x^3 + 3ax^2 + 2x$$

$x = 2$ is a double root

$$\therefore f(2) = 0 \quad \& \quad f'(2) = 0$$

$$\text{i.e. } 16 + 8a + 4 + b = 0 \quad \& \quad 32 + 12a + 4 = 0$$

$$20 + 8a + b = 0 \qquad 12a = -36$$

$$20 - 24 + b = 0 \quad \swarrow \quad a = -3$$

$$b = 4$$

$$\therefore a = -3, b = 4$$

ii) $f(x) = x^4 - 3x^3 + x^2 + 4$

$$= (x-2)^2 \cdot \phi(x)$$

$$= (x-2)^2 (x^2 + x + 1)$$

c. $x^3 - 5x^2 + 5 = 0$

i) let $y = x - 1 \Rightarrow x = y + 1$

$$\text{E: } (y+1)^3 - 5(y+1)^2 + 5 = 0$$

$$y^3 - 2y^2 - 7y + 1 = 0$$

$$(\text{or}) \quad x^3 - 2x^2 - 7x + 1 = 0$$

ii) let $y = x^2 \Rightarrow x = \sqrt{y}$

$$\text{E: } (\sqrt{y})^3 - 5(\sqrt{y})^2 + 5 = 0$$

$$y\sqrt{y} - 5y + 5 = 0$$

$$y\sqrt{y} = 5y - 5 \quad \text{squaring both sides}$$

$$y^3 = 25y^2 - 50y + 25$$

$$y^3 - 25y^2 + 50y - 25 = 0$$

$$(\text{or}) \quad x^3 - 25x^2 + 50x - 25 = 0$$

iii) If $\alpha, \beta \neq \gamma$ are roots of $x^3 - 5x^2 + 5 = 0$, then

$$\alpha^3 - 5\alpha^2 + 5 = 0 \quad \text{--- (1)}$$

$$\beta^3 - 5\beta^2 + 5 = 0 \quad \text{--- (2)}$$

$$\gamma^3 - 5\gamma^2 + 5 = 0 \quad \text{--- (3)}$$

Adding
(1) + (2) + (3)

$$\alpha^3 + \beta^3 + \gamma^3 - 5(\alpha^2 + \beta^2 + \gamma^2) + 15 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = 5(\alpha^2 + \beta^2 + \gamma^2) - 15$$

$$= 5(25) - 15$$

$$= 110$$

sum of roots
from (ii) $\alpha^2 + \beta^2 + \gamma^2 = 25$

Q7 (cont)

d. $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$

i) Polynomials with real coefficients have complex roots in conjugate pairs

$\therefore 2 - 3i$ is also a solution.

ii) $2 + 3i, 2 - 3i$

let $z = 2 + 3i$

$\bar{z} = 2 - 3i$

$z + \bar{z} = 4$; $z\bar{z} = 13$

$\therefore (x - z)(x - \bar{z}) = x^2 - (z + \bar{z})x + z\bar{z}$
 $= x^2 - 4x + 13$

$$\begin{array}{r} x^2 - 2x + 5 \\ x^2 - 4x + 13 \overline{) x^4 - 6x^3 + 26x^2 - 46x + 65} \\ \underline{- x^4 + 4x^3 - 13x^2} \\ - 2x^3 + 13x^2 - 46x \\ \underline{- -2x^3 + 8x^2 - 26x} \\ 5x^2 - 20x + 65 \\ \underline{- 5x^2 - 20x + 65} \\ 0 \end{array}$$

$x^2 - 2x + 5 = 0$

$x^2 - 2x = -5$

$(x-1)^2 = -4$

$x-1 = \pm \sqrt{4i^2}$

$x-1 = \pm 2i$

$x = 1 + 2i$

$\therefore$ Roots : $2 + 3i, 2 - 3i, 1 + 2i, 1 - 2i$

Question 8

a. i) $P(x_0, y_0)$; $\frac{x^2}{25} + \frac{y^2}{9} = 1$

$$a = 5, b = 3$$

$$b^2 = a^2 (1 - e^2)$$

$$9 = 25 (1 - e^2)$$

$$e = \frac{4}{5}$$

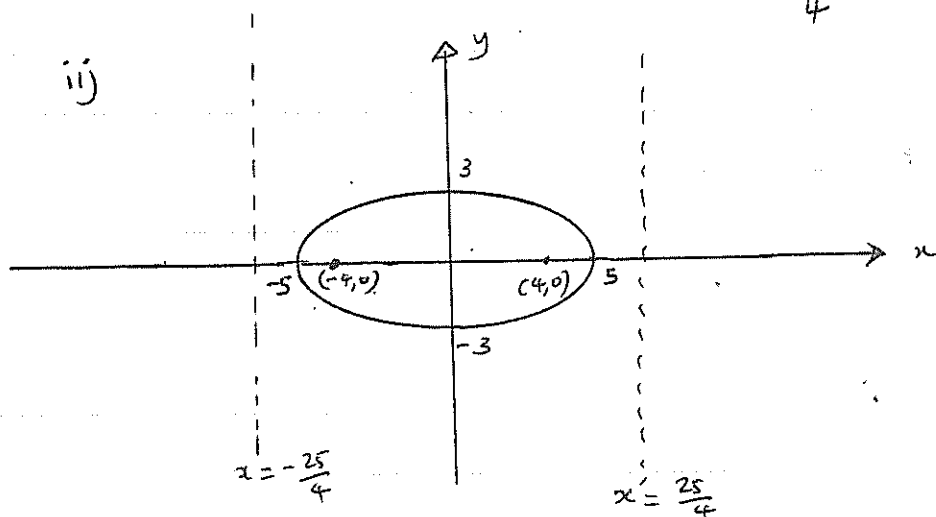
Foci: $(\pm ae, 0)$

$$= (\pm 4, 0)$$

Directrices:

$$x = \pm \frac{a}{e}$$

$$x = \pm \frac{25}{4}$$



iii) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

By implicit differentiation,

$$\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-9x}{25y}$$

At $P(x_0, y_0)$, $m_{\text{tangent}} = \frac{-9x_0}{25y_0}$

$E_{\text{tangent}}: y - y_0 = \frac{-9x_0}{25y_0} (x - x_0)$

$$25y_0 y - 25y_0^2 = -9x_0 x + 9x_0^2$$

$$9x_0 x + 25y_0 y = 9x_0^2 + 25y_0^2 \quad (\div 25)$$

$$\frac{xx_0}{25} + \frac{yy_0}{9} = \frac{x_0^2}{25} + \frac{y_0^2}{9}$$

$\therefore E_{\text{tangent}}: \frac{xx_0}{25} + \frac{yy_0}{9} = 1$

(since (x_0, y_0) lies on the ellipse $\frac{x_0^2}{25} + \frac{y_0^2}{9} = 1$)

Q8 cont

iv) Q on directrix $\therefore x = \frac{25}{4}$

Q lies on tangent

$$\therefore \frac{\frac{25}{4}x_0}{25} + \frac{yy_0}{9} = 1$$

$$\frac{yy_0}{9} = 1 - \frac{x_0}{4}$$

$$\frac{yy_0}{9} = \frac{4-x_0}{4}$$

$$y = \frac{9(4-x_0)}{4y_0}$$

$$\therefore Q\left(\frac{25}{4}, \frac{9(4-x_0)}{4y_0}\right)$$

$$m_{QS} = \frac{9(4-x_0) - 0}{4y_0}$$

$$m_{PS} = \frac{y_0 - 0}{x_0 - 4}$$

$$\frac{\frac{25}{4} - 4}{4}$$

$$= \frac{y_0}{x_0 - 4}$$

$$= \frac{4-x_0}{y_0}$$

$$m_{QS} \cdot m_{PS} = -1$$

$\therefore \angle PSQ$ is a right angle.

b. i) The equation $|z+3| + |z-3| = 10$ describes a point z moving in the complex plane such that the sum of its distances to the points $(3,0)$ & $(-3,0)$ is a constant (10)

$$2a = 10 \Rightarrow a = 5$$

i.e. an ellipse

Foci $S(3,0)$, $S'(-3,0)$

$$ae = 3 \therefore e = \frac{3}{5} ; b^2 = 5^2 \left(1 - \left(\frac{3}{5}\right)^2\right)$$

$$b = 4$$

$$\text{Equation } \frac{x^2}{25} + \frac{y^2}{16} = 1$$

Question 9

$$\frac{x^2}{4} - \frac{y^2}{12} = 1$$

(i) $a^2 = 4, b^2 = 12$

$$b^2 = a^2(e^2 - 1)$$

$$e^2 = 4$$

$$e = 2$$

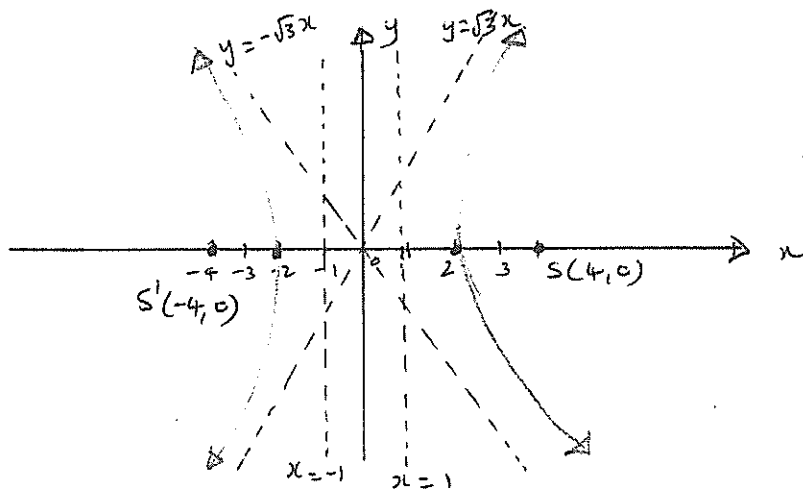
Foci: $(\pm ae, 0)$
 $= (\pm 4, 0)$

Directrices: $x = \pm \frac{a}{e}$
 $x = \pm 1$

Asymptotes: $y = \pm \frac{b}{a}x$

$$= \pm \frac{\sqrt{12}}{\sqrt{4}}x$$

$$y = \pm \sqrt{3}x$$



ii) $P(4, 6); Q(14, 24)$

$$\therefore M_{PQ} = (9, 15)$$

Tangent at P: $\frac{4x}{4} - \frac{6y}{12} = 1$

$$x - \frac{y}{2} = 1$$

$$2x - y = 2$$

Tangent at Q: $\frac{14x}{4} - \frac{24y}{12} = 1$

$$\frac{7x}{2} - 2y = 1$$

$$7x - 4y = 2$$

Tangents intersect at R

Solving ① & ② simultaneously,

$$2x - y = 2 \text{ --- ① } (\times 4)$$

$$7x - 4y = 2 \text{ --- ②}$$

$$6x - 4y = 8 \text{ --- ③}$$

$$\text{③} - \text{① } x = 6$$

$$2(6) - y = 2$$

$$y = 10$$

$$\therefore R(6, 10)$$

$$\text{Gradient of OM} = \frac{15}{9} = \frac{5}{3}$$

$$\text{Gradient of OR} = \frac{10}{6} = \frac{5}{3}$$

$\therefore R, O$ and M are collinear.

Question 10

a. $P(2p, \frac{2}{p})$; $Q(2q, \frac{2}{q})$; $xy = 4$

$$\begin{aligned} \text{i) } m_{PQ} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{\frac{2}{q} - \frac{2}{p}}{2q - 2p} \\ &= \frac{2p - 2q}{pq} \\ &= \frac{-2(p - q)}{-2(p - q)} \\ &= -\frac{1}{pq} \end{aligned}$$

$$E_{PQ} : y - \frac{2}{p} = -\frac{1}{pq}(x - 2p)$$

$$\begin{aligned} pqy - 2q &= -x + 2p \\ x + pqy &= 2(p + q) \end{aligned}$$

ii) PQ passes through $R(4, 2)$

$$\therefore 4 + pq(2) = 2(p + q)$$

$$2(2 + pq) = 2(p + q)$$

$$2 + pq = p + q$$

$$pq = p + q - 2$$

iii) M is the midpoint of PQ

$$\therefore M = \left(\frac{p+q}{2}, \frac{\frac{2}{p} + \frac{2}{q}}{2} \right)$$

$$x = \frac{p+q}{2} ; y = \frac{\frac{2}{p} + \frac{2}{q}}{2}$$

$$\therefore y = \frac{x}{pq}$$

$$= \frac{x}{p+q-2} \quad \begin{matrix} \text{(from ii)} \\ (x = \frac{p+q}{2}) \end{matrix}$$

$$y = \frac{x}{x-2} \text{ is the locus of M.}$$

Q10 (cont)

b. $P(cP, \frac{c}{P})$; $Q(cq, \frac{c}{q})$; $xy = c^2$

i) At $(ct, \frac{c}{t})$

$$y = \frac{c^2}{x} = c^2 x^{-1}$$

$$\frac{dy}{dx} = \frac{-c^2}{x^2}$$

$$m_{\text{tangent}} = \frac{-c^2}{c^2 t^2} \\ = -\frac{1}{t^2}$$

$$E_{\text{tangent}} : y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

$$yt^2 - ct = -x + ct$$

$$\therefore x + t^2 y = 2ct$$

ii) $E_{\text{tangent at } P} : x + p^2 y = 2cp$ — (1)

$$E_{\text{tangent at } Q} : x + q^2 y = 2cq$$
 — (2)

Tangents at P and Q intersect where

$$(p^2 - q^2)y = 2c(p - q)$$

$$(p - q)(p + q)y = 2c(p - q)$$

$$y = \frac{2c}{p + q}$$

substituting into (1)

$$x + p^2 \left(\frac{2c}{p + q} \right) = 2cp$$

$$x(p + q) + 2cp^2 = 2cp(p + q)$$

$$x(p + q) + 2cp^2 = 2cp^2 + 2cpq$$

$$x(p + q) = 2cpq$$

$$x = \frac{2cpq}{p + q}$$

$$\therefore X = \frac{2cpq}{p + q} ; Y = \frac{2c}{p + q} \quad (p \neq q)$$

Q10 cont

$$\text{iii) } p^2 + q^2 = (p+q)^2 - 2pq \\ (p^2 + q^2 = 2)$$

$$\therefore (p+q)^2 = 2 + 2pq \\ (p+q)^2 = 2(1+pq)$$

A + R

$$X = \frac{2cpq}{p+q} ; Y = \frac{2c}{p+q}$$

$$pq = \frac{X(p+q)}{2c}$$

$$pq = \frac{X}{Y}$$

$$p+q = \frac{2c}{Y}$$

$\therefore$ the locus of R has equation

$$\left(\frac{2c}{Y}\right)^2 = 2\left(1 + \frac{X}{Y}\right)$$

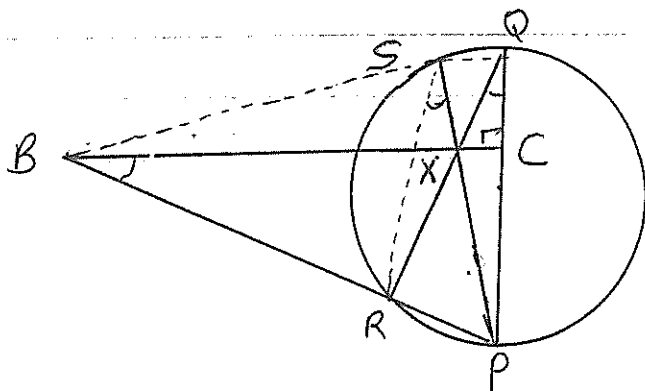
$$\frac{4c^2}{Y^2} = 2\left(1 + \frac{X}{Y}\right)$$

$$2c^2 = Y^2\left(1 + \frac{X}{Y}\right)$$

$$2c^2 = Y^2 + XY$$

$$\text{or } Y^2 + XY = 2c^2$$

Question 11 (8 marks)



- i) In ΔBCP and ΔPRQ ,
 $\angle PRQ = 90^\circ$ ($\angle$ in a semi-circle)
 $\angle BPC$ is common
 $\therefore \Delta BCP \equiv \Delta PRQ$ (equiangular)
 $\therefore \angle PBC = \angle PQR$ (matching $\angle$ s in $\equiv \Delta$ s)
- ii) Join SR
 $\angle RSP = \angle RQP$ ($\angle$ s at the circumference standing on same arc RP)
 $\angle RBC = \angle RSP$ (both equal to $\angle RQP$)
 $\therefore SBRX$ is a cyclic quadrilateral
 (Interval RX subtends equal $\angle$ s on same side)
- iii) Join BS and SQ
 $\angle PSQ = 90^\circ$ ($\angle$ in a semi-circle)
 $\angle PRX = \angle BSX$ (exterior $\angle$ of cyclic quadrilateral = interior opposite $\angle$)
 Since $\angle PRX = 90^\circ$, $\angle BSX = 90^\circ$
 $\angle BSQ = \angle BSX + \angle XSQ$
 $= 90^\circ + 90^\circ$
 $= 180^\circ$
 $\therefore \angle BSQ$ is a straight $\angle$ and
 B, S and Q are collinear.